MILITARY SPECIFICATION

FUZE, MTSQ, M548 ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Fuze, MTSQ, M548 Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-60412	- Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes, General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Test for
MIL-STD-1167	- Ammunition Data Card
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables and Percent Defective

DRAWINGS

Armament Command

F8864492	- Box, Ammunition, Metal for Artillery Type and Rocket Fuzes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in approving this document should be addressed to: Commander, US Army Armament Research & Development Command, ATTN: DRDAR-TST-S, Dover, NJ 0780 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1390

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2.1 DRAWINGS (Cont'd)

F10520638	- Fuze, MTSQ, M548 Assembly
IEL-10520638	- Inspection Equipment List for Fuze, MTSQ, M548 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B46.1	- Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards, and specifications except as provided in 6.1.

3.2 Dynamic regulation.-The movement subassembly shall be regulated to the frequency specified on the applicable drawing. The timing error between 2 and 100 seconds shall not exceed 0.18% while spinning at the specified revolutions per minute (rpm).

3.3 Environmental.

3.3.1 Jolt.-The fuze shall comply with MIL-STD-331, Test No. 101.1, except as specified herein (see 4.4.14.1).

3.3.2 Jumble.-The fuze shall comply with MIL-STD-331, Test No. 102.1.

3.3.3 Forty-foot drop.-The fuze shall comply with MIL-STD-331, Test No. 103.

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3.3.4 Waterproofness. The fuze shall comply with MIL-STD-331, Test No. 108, except that the duration of immersion shall be five (5) minutes and the pressure shall be three (3) psi gauge.

3.3.5 Transportation-Vibration. The fuze shall comply with MIL-STD-331, Test No. 104.

3.4 Ballistics. (see 6.3.4)

3.4.1 The fuze shall not function prematurely. (see 6.3.1)

3.4.2 Target impact nonfunction distance. The fuze shall not initiate detonation of the round upon impact element function when fired in the 105MM Howitzer, Charge 7 (1550 feet per second, fps) with the fuze safe "S" on a target 150 plus 5 feet from the muzzle of the weapon.

3.4.3 Target impact function distance. The fuze shall initiate detonation of the round upon impact when fired in the 105MM Howitzer, Charge 7 (1550 fps) with the fuze set safe "S" on a target 400 minus 5 feet from the muzzle of the weapon.

3.4.4 Ground or water impact function. The fuze shall initiate detonation of the round upon impact when fired as follows:

Weapon	Charge	Impact Angle	Setting
105MM Howitzer	1 (650 fps)	45° minimum	Safe "S"
105MM Howitzer	7 (1550 fps)	45° minimum	Safe "S"

3.4.5 Time-to-burst (Ambient temperature). The fuze shall initiate detonation of the round during flight within the time limits specified in Table I, when fired as follows:

Weapon	Charge	Setting
155MM Howitzer	1 (680 fps)	3 seconds
105MM Howitzer	7 (1550 fps)	60 seconds

TABLE I

Weapon	Charge	Fuze Setting	Individual Round Function Time Limits (seconds)	
155MM How.	1	3 seconds	UL	3.47
			LL	2.88
105MM How.	7	60 seconds	UL	59.60
			LL	57.70

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3.4.6 Excess pressure. The fuze shall initiate detonation of the round furing flight when fired in the 105MM Howitzer at 112% of the normal charge 7 pressure with the fuze set at 60 seconds.

3.4.7 Low temperature. The fuze temperature conditioned at minus 65 to minus 70 degrees Farenheit (F) shall initiate detonation of the round during flight when fired as specified in 3.4.5 (see 6.4).

3.4.8 High temperature. The fuze temperature conditioned at plus 145 to plus 150 degrees Farenheit (F) shall initiate detonation of the round during flight when fired as specified in 3.4.5 (see 6.4).

3.4.9 High and Low Temperature conditioning. For the high and low temperature firings, only the fuze shall be temperature conditioned. The elapsed time from the conditioning chamber to the firing in the weapon shall not exceed five (5) minutes. Reconditioning is not permitted.

3.5 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions or any similar defect.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing.

3.5.3 Burrs. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.4 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter. The methods and agents used in processing and cleaning shall not adversely affect the finish or contaminate the parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions. Reference shall be made to MIL-STD-109 to define quality assurance terms used.

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4.1.2 Lot formation. Each inspection lot of fuzes shall be comprised of components (except explosive components) each produced by one manufacturer, by one process in accordance with the same drawing and revision and specification and revision, as applicable, and assembled on one assembly line. Each explosive component within a lot of assembled fuzes shall be limited to one manufacturer and one interfix number.

4.2 First article sample. When required, the first article sample shall be as specified in the contract and shall consist of the following:

340 or 360 (aggregate) (as applicable, see 4.2.2.3) component parts as delineated in 4.2.2
450 (aggregate) subassemblies as delineated in 4.2.3
147 (aggregate) fuze assemblies (Drawing F10520638) as delineated in 4.2.5 and 4.2.6

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1 Examination and test. After inspection by the contractor and provisional acceptance at source by the Government, 125 fuze assemblies shall be forwarded to a designated proving ground as specified in the contract. Three hundred and forty (340) or 360 component parts, 450 subassemblies and 22 fuze assemblies shall be inspected, as provided herein, at a Government laboratory or other facility as specified in the contract.

4.2.2 Components.

4.2.2.1 Twenty (20) of each of the (16) components shall be tested as in 4.3.3.1 (Hardness). The firing arms shall also be tested as in 4.3.3.2 (Bend).

4.2.2.2 Twenty (20) of each component (1) shall be tested as in 4.3.3.3 (Load).

4.2.2.3 Twenty (20) of each component (1), if applicable, shall be tested as in 4.3.3.4 (Force).

4.2.3 Subassemblies.

4.2.3.1 Twenty-five (25) of each of the (16) subassemblies shall be tested as in 4.3.3.6.1 (Static load).

4.2.3.2 Twenty-five (25) of each of the (2) subassemblies shall be tested as in 4.3.3.6.2 (Torque).

4.2.4 Acceptance criteria. Failure of any component or subassembly to comply with the requirements of the applicable drawing for the above characteristics shall be cause for rejection of the first article sample.

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4.2.5 Fuze assemblies. Twenty-two (22) fuze assemblies, less closing cup assemblies, shall be subjected to the following MIL-STD tests, with the fuzes set on safe "S".

Test	Quantity	Requirement	Test Method
Jolt	4	3.3.1	4.4.14.1
Jumble	4	3.3.2	4.4.14.2
Forty-foot drop	5	3.3.3	4.4.14.3
Waterproofness	4	3.3.4	4.4.14.4
Transportation-Vibration	5	3.3.5	4.4.14.5

4.2.5.1 Acceptance criteria. The criteria of the applicable MIL-STD shall apply as determined by disassembly and examination in accordance with the procedures in 4.4.14. Where the accept/reject decision is based on engineering judgement, concurrence by the responsible technical agency must be obtained.

NOTE: Evidence of water entry through the D.A.M. flash hole or setback housing hole does not constitute reason for rejection.

4.2.6 Ballistics. One hundred and twenty-five (125) fuze assemblies of the first article sample shall be subjected to the following ballistic test phases.

Phases	Quantity	Requirement	Test Method
Target Impact Nonfunction Distance (Ambient Temperature)	20	3.4.1 & 2	4.4.15.1
Target Impact Function Distance (Ambient Temperature)	10	3.4.1 & 3	4.4.15.2
Ground or Water Impact Function (Ambient Temperature)			
105MM Howitzer Charge 1	5	3.4.1 & 4	4.4.15.3
105MM Howitzer Charge 7	5	3.4.1 & 4	4.4.15.3
Time-to-burst (Ambient Temperature)			
155MM Howitzer Charge 1	20	3.4.1 & 5	4.4.15.4
105MM Howitzer Charge 7	20	3.4.1 & 5	4.4.15.4
Excess pressure (Ambient Temperature)	25	3.4.1 & 6	4.4.15.5
Low Temperature (-65° to -70°F)			
155MM Howitzer Charge 1	5	3.4.1 & 7	4.4.15.6
105MM Howitzer Charge 7	5	3.4.1 & 7	4.4.15.6
High Temperature (+145° to +150°F)			
155MM Howitzer Charge 1	5	3.4.1 & 8	4.4.15.6
105MM Howitzer Charge 7	5	3.4.1 & 8	4.4.15.6

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4.2.6.1 Acceptance criteria. The occurrence of any of the following shall be cause for rejection of the first article sample.

a. A premature function in any phase (see 6.3.1).

b. Function of one or more rounds upon target impact in the Target Impact Nonfunction Distance Phase.

c. Failure of more than one round to function in the Target Impact Function Distance Phase.

d. Failure of more than one round to function in the Ground or Water Impact Function Phase.

e. Any of the following in the ambient Time-to-Burst Phase at either time setting.

(1) More than one time failure (6.3.2).

(2) More than one outlier-offtime (6.3.3).

(3) Failure to meet the limits specified in Table I (see 3.4.5) by an estimated lot percent defective greater than the appropriate value given in Table II (see 4.4.14.4.1).

f. More than two time failures (6.3.2) in the Excess Pressure Phase.

g. More than two time failures (6.3.2) in either of the Environmental Phases (Low Temperature, High Temperature).

TABLE II

3 Second Setting

Sample Size	Allowable Percent Defective
20	2.95
19	2.96
18	2.97
17	2.99
16	3.01
15	3.05

60 Second Setting

20	4.00
19	4.01
18	4.03
17	4.06
16	4.12
15	4.20

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4.3 Inspection provisions.

4.3.1 Submission of product. The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major defects shall be performed on a single defect basis in accordance with MIL-STD-105. For each classification of defects contained herein, the AQL given shall be applied to each major defect listed. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4x) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

4.3.2.1 Cup, Closing, Drawing B 8595508, covering a detail of Dwg. F10520638.

Defect	Method of Inspection
CRITICAL	
1. Split, cut or crack	Visual
MAJOR AQL = 1.0%	
101. Thickness of base, incorrect	Gage
102. Length under flange, incorrect	Gage
103. Outside diameter exceeds max.	Gage
104. Score, wrinkle or dent	Visual
105. Foreign matter	Visual

4.3.2.2 Closing Cup Assembly, Drawing B8595506, covering a detail of Drawing F10520638.

CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Explosive above flush	Gage
102. Explosive below flush exceeds max.	Gage
103. Cup damaged	Visual

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4.3.2.3 Pin, Setting, Drawing B10520656 covering a detail of Drawing F10520638.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Large diameter below min. allowable	Gage
102. Small diameter, incorrect	Gage
103. Radius at base of small diameter	Visual

4.3.2.4 Plate No. 1, First Lamina, Drawing F10523181 covering a detail of Drawing F10520638.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter of holes M and 20, incorrect	Gage
102. Location of holes M and 20, incorrect	Gage
103. Surface finish of holes M and 20, improper	Visual
104. Metal burred or damaged	Visual
105. Chamfer on holes M and 20, missing or incorrect	Visual

4.3.2.5 Plate No. 2, Second Lamina, Drawing F8595968, covering a detail of Drawing F10520638.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Diameter of holes M and 20, incorrect	Gage
102. Location of holes M and 20, incorrect	Gage
103. Surface finish of holes M and 20, improper	Visual
104. Metal burred or damaged	Visual
105. Chamfer on holes M and 20, missing or incorrect	Visual

4.3.2.6 Plate No. 2, First Lamina, Drawing F8595967, covering a detail of Drawing F10520638.

CRITICAL	
None defined	

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Flatness incorrect	Gage
102. Metal burred or damaged	Visual

4.3.2.7 Slide, Pallet Lock, Drawing C10521169, covering a detail of Drawing F10520638.

CRITICAL

1. Large radius on extension above max.	Gage

MAJOR

None defined

4.3.2.8 Gear, Escapement, Drawing D10521109, covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Tooth profile, incorrect	Gage
102. Surface finish, improper	Visual

4.3.2.9 Plate No. 7, Drawing F10520737, covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Outside diameter, incorrect	Gage
102. Surface finish on outside diameter, improper	Gage
103. Burr	Visual

4.3.2.10 Escapement Gear and Pinion Assembly, Drawing C10521112, covering a detail of Drawing F10520638.

CRITICAL

None defined

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4.3.2.10 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Concentricity of gear, incorrect	Gage
102. Perpendicularity of gear, incorrect	Gage
103. Burr	Visual

4.3.2.11 Escapement Lever Assembly, Drawing F8593015, covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Spring burred or damaged	Visual
102. Improper assembly	Visual

4.3.2.12 Arbor, Escapement, Drawing D10534857, covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Perpendicularity of lever seat, incorrect	Gage
102. Diameter of pivots, incorrect	Gage
103. Metal burred or damaged	Visual

4.3.2.13 Head, Drawing F10520719, covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%	
101. Profile of external surface incorrect	Gage
102. Threads damaged or not full	Visual

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4.3.2.14 Pin, Firing, Drawing C10521154, covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Profile of striker point, incorrect	Gage
102.	Surface finish on shoulder angle, improper	Visual

4.3.2.15 Pin, Firing, Drawing C10542841, covering a detail of Drawing F10520638.

CRITICAL

1.	Diameter of neck area exceeds max.	Gage
2.	Length of striker exceeds max.	Gage
3.	Diameter of body over the striker less than min.	Gage
4.	Corner radius of body shoulder (over striker) exceeds max.	Gage

MAJOR AQL = 1.0%

101.	Diameter of neck area less than min.	Gage
102.	Diameter of body over the striker exceeds max.	Gage
103.	Profile of striker point, incorrect	Gage
104.	Runout of striker diameter, incorrect	Gage

4.3.2.16 Plug, Closing, Drawing D10530733, covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Location of locating pin hole, incorrect	Gage

4.3.2.17 Cap, Drawing F10520711, covering a detail of Drawing F10520638.

CRITICAL

1.	Letter "S" not over its proper line	Visual

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4.3.2.17 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 1.0%	
101. Profile of external surface, incorrect	Gage
102. Diameter of large cavity, incorrect	Gage
103. Cumulative error of time scale exceeds max. (to be measured from zero line to 3, 30, 60 and 90 second lines)	Gage
104. Diameter of small hole between spotface exceeds max.	Gage
105. Location of numerals, incorrect	Visual
106. Spot face, missing	Visual
107. Surface finish of large cavity, inproper	Visual
108. Threads damaged or not full	Visual

4.3.2.18 Body (Fuze),Drawing F10520709 covering a detail of Drawing F10520638.

CRITICAL	
1. Minor diameter of internal thread exceeds max.	Gage
2. Internal threads not full to undercut	Gage
MAJOR AQL = 1.0%	
101. Pitch diameter of internal thread, improper	Gage
102. Runout of minor diameter of internal thread, incorrect	Gage
103. Major diameter of external thread, improper	Gage
104. Pitch diameter of external thread, improper	Gage
105. Diameter of internal flange (mid-section), incorrect	Gage
106. Runout of internal flange (mid-section), incorrect	Gage
107. Profile of external surface, incorrect	Gage
108. Threads damaged or not full	Visual

4.3.2.19 Base, Movement, DrawingF10520728 covering a detail of Dwg. F10520638.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Location of closing plug locating hole, incorrect	Gage

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4.3.2.20 Ring, Locking, Drawing C10520731 covering a detail of Dwg. F10520638

Defect	Method of Inspection
CRITICAL	
1. Large inside diameter exceeds max.	Gage
2. Major diameter of thread below min.	Gage
MAJOR AQL = 1.0%	
101. Pitch diameter of thread, improper	Gage

4.3.2.21 Body and Cap Assembly, Drawing F10520720 covering a detail of Drawing F10520638.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Wave washer missing	Visual
102. Snap ring missing	Visual
103. Threads damaged	Visual
104. Improper or incomplete assembly	Visual

4.3.2.22 Ring, Reinforcing, Drawing B10520736 covering a detail of Dwg. F10520638

CRITICAL	
None Defined	
MAJOR AQL = 1.0%	
1. Outside diameter below min.	Gage

4.3.2.23 Escapement Wheel and Pinion Assembly, Drawing D11743967 Covering a detail of Drawing F10520638.

CRITICAL	
None defined	
MAJOR AQL = 1.0%	
101. Circular runout of escape wheel and pinion, incorrect	Gage

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4.3.2.23 (Cont'd)

	Defect	Method of Inspection
102.	Perpendicularity of escape wheel and pinion, incorrect	Gage
103.	Broken or missing teeth, improper assembly	Visual
104.	Escape wheel not seated properly (see 6.6)	Visual
105.	Metal defective	Visual

4.3.2.24 No. 1 Gear and Pinion Assembly, Drawing D10523167 covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Circular runout of gear and pinion, incorrect	Gage
102.	Perpendicularity of gear and pinion, incorrect	Gage
103.	Broken or missing teeth, improper assembly	Visual
104.	Gear not seated properly (see 6.6)	Visual
105.	Metal defective	Visual

4.3.2.25 Wheel, Escape, Drawing C8595539 covering a detail of Dwg. F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Escape wheel profile, incorrect	Gage
102.	Metal defective	Visual

4.3.2.26 Pallet Assembly, Drawing D11743962 covering a detail of Dwg. F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Perpendicularity of pallet, improper	Gage
102.	Broken, deformed, or bent assembly	Visual
103.	Pallet not seated properly (see 6.6)	Visual
104.	Metal defective	Visual

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4.3.2.27 Rotor Assembly, Drawing D10551356 or Drawing D11744013 covering a detail of Drawing F10520638.

CRITICAL

None defined

MAJOR AQL = 1.0%

101.	Circular runout of rotor assembly, incorrect	Gage
102.	Perpendicularity of rotor assembly, incorrect	Gage
103.	Broken or missing teeth, improper assembly	Visual
104.	Metal Defective	Visual

4.3.2.28 Body, Delay Arming Mechanism, (D.A.M.), Drawing F10520703 covering a detail of Drawing F10520638 (NOTE: Inspection of critical and major dimensions may be performed prior to finish coating.)

CRITICAL

1.	Major diameter of large external thread less than min.	Gage
2.	Large straight inside diameter exceeds max.	Gage

MAJOR AQL = 1.0%

101.	Pitch diameter of large external thread, improper	Gage
102.	Major diameter of small external thread, improper	Gage
103.	Pitch diameter of small external thread, improper	Gage
104.	Depth of setback housing hole, incorrect	Gage
105.	Threads damaged or not full	Visual
106.	Metal defective	Visual

4.3.2.29 Detonator Screw Assembly, Drawing C10520294, covering a detail of Drawing F10520638.

CRITICAL

1.	Detonator extends above flush of detonator screw	Visual

MAJOR AQL = 0.40%

101.	Green end of detonator not at bottom (thread end)	Visual
102.	Crimping inadequate	Visual
103.	Threads damaged or not full	Visual

4.3.2.30 Detent Safety Mechanism Assembly, Dwg.F10520250, covering a detail of Drawing F10520638.

CRITICAL

1.	Length from top of detents to bottom of housing less than min.	Gage

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4.3.2.30 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.40%	
101. Improper or incomplete assembly	Manual/Visual
102. Staking of cover to housing inadequate	Visual
103. Foreign matter	Visual

4.3.2.31 Cap Assembly, Drawing D10520710 covering a detail of Dwg. F10520638.

CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Protrusion of pin, incorrect	Gage
102. Edge of hammer spring not contacting surfact "A"	Visual
103. Threads damaged	Visual
104. Burr	Visual

4.3.2.32 Firing Arm Assembly, Drawing C10521115, covering a detail of Drawing F10520638.

CRITICAL	
1. Fracture of upright bend of firing arm	Visual
2. Firing arm not staked to shaft	Visual
MAJOR AQL = 0.40%	
101. Insecure weight	Manual/Visual
102. Improper or incomplete assembly	Visual
103. Metal burred or damaged	Visual

4.3.2.33 Timing Disc Assembly, Drawing D10520653 covering a detail of Drawing F10520638.

CRITICAL	
1. Safety disc tabs not engaged with timing disc bushing slots	Visual
MAJOR AQL = 0.40%	
101. Safety disc radius below periphery of timing disc	Visual
102. Improper or incomplete assembly	Visual

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4.3.2.34 Movement Subassembly (after dynamic regulation), Drawing F10520716 covering a detail of Drawing F10520638.

	Defect	Method of Inspection
CRITICAL		
1.	Firing pin fires on "not fire" position of firing arm	Gage
2.	Extension of firing arm upright above top plate less than minimum (with firing arm shaft shoulder bottomed on plate No. 2)	Gage
3.	Escapement gear not locked in sufficiently by lever pallet (check after rewind and removal of backlash)	Visual
MAJOR	AQL = 0.40%	
101.	Firing pin does not fire on "must fire" position of firing arm	Gage
102.	Firing pin protrusion less than min. in fired position	Gage
103.	Extension of firing arm upright above top plate exceeds maximum (with firing arm shaft shouldered against top plate)	Gage
104.	Firing pin protrudes	Manual/Visual
105.	Staking of assembly screws inadequate (check prior to dynamic regulation)	Visual
106.	Arbor center hole missing	Visual
107.	Escapement lever not locked on proper side of pallet lock slides	Visual
108.	Firing pin striker point damaged	Visual
109.	Foreign matter	Visual
110.	Firing arm shaft spring protrudes	Visual

4.3.2.35 Movement Assembly, Drawing F10520715 covering a detail of Dwg. F10520638.

CRITICAL		
1.	Drop off edge of firing arm outside of lines on safety disc	Visual
2.	Timing disc bushing loose on knurl of arbor (check prior to assembly of locknut)	Manual/Visual
3.	Timing disc retainer insecure, missing or fractured at bend	Manual/Visual
4.	Distance from top of top plate to underside of retainer exceeds max.	Gage

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4.3.2.35 (Cont'd)

Defect	Method of Inspection
MAJOR AQL - 0.40%	
101. Clearance between No. 7 plate and safety disc below min. allowable	Gage
102. Interference of the timing disc retainer the safety disc	Visual
103. Staking of assembly screws inadequate	Visual
104. Foreign matter	Visual
105. Improper or incomplete assembly	Visual

4.3.2.36 Body, Movement and Cap Assembly, Drawing F10520708 covering a detail of Drawing F10520638.

CRITICAL	
1. Setting pin not engaged in timing disc tab slot	Visual
2. Timing disc tab not contacting setting pin shoulder	Visual
MAJOR AQL = 0.40%	
101. Movement not properly secured by locking ring	Visual
102. Burr	Visual

4.3.2.37 Point Detonating Assembly, Drawing D10520718, covering a detail of Crawing F10520638.

CRITICAL	
1. Detent safety mechanism assembly missing (to be checked after staking)	Visual
MAJOR AQL = 0.40%	
101. Improper or incomplete assembly	Visual
102. Crimping of closing disc inadequate	Visual
103. Threads damaged	Visual
104. Burr	Visual
105. Foreign matter	Visual

4.3.2.38 Setback Mechanism Assembly, Drawing D10551357, covering a detail of Drawing F10520638.

CRITICAL

None defined

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4.3.2.38 (Cont'd)

Defect	Method of Inspection
MAJOR AQL = 0.40%	
101. Crimp missing or inadequate	Visual

4.3.2.39 Delay Arming Mechanism Assembly, Drawing F10520702 covering a detail of Drawing F10520638.

CRITICAL	
None defined	
MAJOR AQL = 0.40%	
101. Length from base of body to top plate including all protrusions exceeds max.	Gage
102. Bulge or protrusion on the base surface	Visual
103. Staking of screw (s) inadequate	Visual
*104. Detonator installed incorrectly (with yellow end at crimp)	Visual
*105. Staking or crimping of detonator inadequate	Visual
*106. Foreign matter	Visual
*107. Insecure, improper or incomplete assembly	Visual
*108. Metal defective	Visual
*109. Threads damaged	Visual
*110. Burrs	Visual
*(to be checked prior to assembly of top plate)	

4.3.2.40 Fuze, MTSO, M548 Assembly, Drawing F10520638

CRITICAL	
1. Reinforcing ring missing (to be checked prior to assembly of D.A.M.)	Visual
2. Fuze not set on safe "S" (to be checked immediately prior to packaging)	Visual
*3. Detent safety mechanism missing	Visual
*4. Detent safety mechanism above flush	Visual
*5. Primer extends above flush	Visual
MAJOR AQL = 0.40%	
101. Total length exceeds max.	Gage

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4.3.2.40 (Cont'd)

	Defect	Method of Inspection
102.	M47 detonator missing (to be checked prior to assembly of closing plug)	Visual
103.	Relay improperly inserted in closing plug (closed end not visible - to be checked prior to assembly of D.A.M.)	Visual
104.	Staking of closing plug screws inadequate	Visual
105.	Threads damaged	Visual
106.	Marking missing, incorrect or illegible	Visual
107.	Dust cover (over lead charge) missing	Visual
108.	Sealing compound missing or inadequate	Visual
109.	Insecure, improper or incomplete assembly	Visual
*110.	Primer missing or installed incorrectly (red end not visible)	Visual
**111.	Closing cup, (lead charge) missing	Visual
**112.	Staking of closing cup assembly missing or inadequate	Visual

*(check prior to assembly of point detonating assembly)
**(check prior to assembly of cover over lead charge)

4.3.2.41 Packaging, packing, and marking. The sample size and acceptance criteria for packaging, packing, and marking shall be in accordance with MIL-P-60412, except as specified in paragraph 5.1.1.

4.3.2.42 Process control audit. Periodic (at least once during each 8-hour shift) unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings. Bending adjustment of the firing arm upright is not permitted at any time.

a. Laminated Plate No. 1 and No. 2 Assembly, Drawing F8595991

(1) Application of lubricant

b. Movement Subassembly, Drawing F10520716

(1) Application of lubricant

(2) Top and bottom assembly screws staked prior to dynamic regulation

(3) Mainspring wound before assembly of flat head screw

(4) Dynamic regulation operation (see 4.4.9.2)

(5) Movement fully wound after dynamic regulation.

(6) Backlash removal operation after final wind

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4.3.2.42 (Cont'd)

c. Escapement Lever Assembly, Drawing F8593015

(1) Escapement lever spring secured

d. Movement Assembly, Drawing F10520715

(1) Locknut torqued prior to timing disc torque test

e. Body and Cap Assembly, Drawing F10520720

(1) Insertion of "O" ring between cap and body

(2) Fuze setting torque

f. D.A.M. Movement Assembly, Drawing F11743960.

(1) Staking of plate screws prior to arming and nonarming tests

(2) Application of lubricant

g. Mainspring Housing and Arbor Assembly, Drawing F10520717

(1) Assembly of #5 gear and arbor assembly for static load and torque tests (see 4.3.3.6)

(2) Staking of lug and push test

h. Fuze, MTSQ, M548 Assembly, Drawing F10520638

(1) Application of lubricant

(2) Application of sealant

(3) Insertion of "O" ring between head and cap

(4) Staking of lead charge operation

(5) Head torque operation

(6) D.A.M. tightening operation

(7) Fuze setting torque - "S" safe

4.3.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness (destructive). Twenty samples from each lot of the following

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4.3.3.1 (Cont'd)

components shall be subjected to the hardness test. Where plating is required, the sample parts tested shall be unplated, where feasible. Failure of any component to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot of components as applicable (see 4.4.1

Component	Drawing Number
Pin, Firing	C10521154
Gear, Escapement	D10521109
Pinion, Escapement	D10521101
Pinion No. 1	D10521119
Pinion No. 2	D10521120
Pinion No. 3	D10521118
Pinion No. 4	C10521123
Wheel, Escape (D.A.M.)	C8595539
Arm, Firing (see 4.3.3.2)	D10521116
Guide, Adjusting	C8594273
Lever, Escapement	F8593023
Spring, Hammer	C10520713
Spring, Timing Disc	C10520645
Washer, Wave	C10520664
Pin, Pallet	B11743965
Ring, Snap	C10520724

4.3.3.2 Bend - Firing arm (destructive). The same 20 firing arms that comply with the hardness requirement shall be subjected to the bend specified on Drawing D10521116. Failure of any firing arm to comply with the requirements shall be cause for rejection of the lot of firing arms (see 4.4.2).

4.3.3.3 Load - Support Cup (destructive). Twenty samples from each lot of support cups shall be subjected to the load specified on Drawing C10521142. Failure of any support cup to comply with the requirement shall be cause for rejection of the lot of support cups (see 4.4.3).

4.3.3.4 Force - Sintered rotor (destructive). Twenty samples from each lot of sintered rotors shall be subjected to the force specified on Drawing F11744012. Failure of any sintered rotor to comply with the requirement shall be cause for rejection of the lot of rotors (see 4.4.4).

4.3.3.5 Compression - Firing pin spring (destructive). Twenty samples from each lot of firing pin springs shall be subjected to the compression specified on Drawing C10521132. Failure of any spring to comply with the requirement shall be cause for rejection of the lot of firing pin springs (see 4.4.5).

4.3.3.6 Static load and torque.

4.3.3.6.1 Static load. Twenty five (25) samples from each lot of the following subassemblies shall be subjected to the static load test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.6).

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4.3.3.6.1 (Cont'd)

Static Load	Drawing Number
Escapement Gear and Pinion Assembly	C10521112
Escapement Lever Assembly	F8593015
Firing Arm Assembly (Test Pin and Arm)	C10521115
No. 1 Gear and Pinion Assembly	D10521102
No. 2 Gear and Pinion Assembly	C10521104
No. 3 Gear and Pinion Assembly	C10521106
No. 4 Gear and Pinion Assembly	C10521111
Detent Safety Mechanism Assembly	F10520250
Pallet Lock Assembly	B10521168
Lever Assembly	B10521148
Escapement Wheel and Pinion Assembly (D.A.M. Assembly)	D11743967
No. 1 Gear and Pinion Assembly (D.A.M. Assembly)	D10523167
Pallet Assembly (Pins and Shaft)	D11743962
*Mainspring Housing and Arbor Assembly (Push on Lug and Push on No. 5 Gear)	F10520717
Closing Plug Loading Assembly	D10520732
** D.A.M. Assembly (Push on Setback Mechanism Assembly)	F10520702

* Only Mainspring Housing F10520729 and Lug C10520739 need be assembled for static load test on lug. Only Arbor C10520648 and Gear No. 5 C10521117 need be assembled for static load test on Gear No. 5. The methods and equipment used to assembly these samples shall be the same ones used in production of the Mainspring Housing and Arbor Assembly.

** Only setback Mechanism Assembly D10551357 need be assembled into D.A.M. Body F10520703 for this static load test. The methods and equipment used to assemble these samples shall be the same ones used in production of the D.A.M. Assembly.

4.3.3.6.2 Torque. Twenty-five (25) samples from each lot of the following subassemblies shall be subjected to the torque test. Failure of any subassembly to comply with the requirement specified on the applicable drawing shall be cause for rejection of the lot of subassemblies, as applicable (see 4.4.6).

Torque	Drawing Number
Escapement Lever Assembly	F8593015
*Mainspring Housing and Arbor Assembly	F10520717

* Only Arbor C10520648 and Gear No. 5 C10521117 need be assembled for torque test. The methods and equipment used to assemble these samples shall be the same ones used in production of the Mainspring Housing and Arbor Assembly.

4.3.3.7 Load - Hammer spring (nondestructive). Hammer springs from each lot of hammer spring assemblies shall be tested to determine compliance with the load requirement of Drawing D10520710. The sample size and criteria for acceptance shall be as specified in MIL-STD-105, using an AQL of 1.0 percent (see 4.4.7).

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4.3.3.8 Load - Setback Mechanism Assembly (nondestructive).-Each setback mechanism assembly produced shall be tested to determine compliance with the load requirements of Drawing D10551357. Failure to comply with the requirement shall be cause for rejection of the setback mechanism assembly (see 4.4.8).

4.3.3.9 Movement Sub-assembly.

4.3.3.9.1 Kinetic Energy of Firing Pin (nondesturctive).-Fifty (50) movement subassemblies from each lot produced shall be tested to determine compliance with the requirement of Drawing F10520716. Failure of any firing pin to comply with the requirement shall be cause for rejection of the lot of movement subassemblies (see 4.4.9.1).

4.3.3.9.2 Dynamic Regulation (nondestructive).-The regulation tape record (100 seconds) of each movement subassembly produced shall be inspected to determir acceptability of the movement. Any movement subassembly which fails to comply with the requirement of 3.2 shall be rejected (see 4.4.9.2).

4.3.3.9.3 Spin Function (nondestructive).-The pallet lock and safety lever assemblies within each movement subassembly produced shall be tested to determine compliance with the function requirement of Drawing F10520716. Failure to comply with the requirement shall be cause for rejection of the movement subassembly (see 4.4.9.3).

4.3.3.9.4 Spin Nonfunction (nondestructive).-Each movement subassembly produced shall be tested to determine compliance with the nonfunction requirement of Drawing F10520716. Failure of the lever lock slides to remain in the locked position shall be cause for rejection of the movement subassmbly (see 4.4.9.4).

4.3.3.10 Movement assembly.

4.3.3.10.1 Timing disc torque (nondestructive).-The timing disc of each movement assembly produced shall be tested to determine compliance with the torque requirement of Drawing F10520715. Failure to comply with the requirement shall be cause for rejection of the movement assembly (see 4.4.10.1).

4.3.3.11 Delay arming mechanism assembly.

4.3.3.11.1 Spin arming less setback assembly (nondestructive).-Each D.A.M. assembly produced shall be tested to determine compliance with the arming requirement of Drawing F10520702. Failure of the rotor to become locked in alignment with flash hole, when spun at the arming spin requirement on the drawing shall be cause for rejection of the assembly (see 4.4.11.1).

4.3.3.11.2 Spin, 3000 + 50 r.p.m. with setback assembly (nondestructive). Each D.A.M. assembly produced shall be tested to determine that the rotor is restricted from moving into the armed position, when spun at 3000 r.p.m. Failure of the setback assembly to restrict movement of the rotor with both spring loaded blocks in the fully opened position, shall be cause for rejection of the assembly (see 4.4.11.2).

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4.3.3.11.3 Spin nonarming with setback assembly - (nondestructive).-Each D.A.M. assembly produced shall be tested to determine compliance with the nonarming requirement of Drawing F10520702. Failure of both spring loaded blocks to remain engaged in the rotor recesses, when spun at the nonarming spin requirement on the drawing shall be cause for rejection of the assembly (see 4.4.11.3).

4.3.3.12 Detent safety mechanism.

4.3.3.12.1 Arming (nondestructive).-Each detent safety mechanism produced shall be tested to determine compliance with the arming requirement of Drawing F10520250. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.12.1).

4.3.3.12.2 Nonarming (nondestructive).-Each detent safety mechanism produced shall be tested to determine compliance with the nonarming requirement of Drawing F10520250. Failure to comply with the requirement shall be cause for rejection of the assembly (see 4.4.12.1).

4.3.3.13 Fuze assembly.

4.3.3.13.1 Setting torque (nondestructive).-Each fuze assembly produced shall be tested to determine compliance with the setting torque requirement of Body and Cap Assembly, Drawing F10520720. Failure to comply with the requirement shall be cause for rejection of the fuze assembly (see 4.4.13.1).

4.3.3.14 Environmental tests (lots) (destructive).-Twenty-two (22) fuze assemblies from each production lot shall be subjected to the following MIL-STD tests, with the fuzes set safe "S".

Test	Quantity	Requirement	Test Method
Jolt	4	3.3.1	4.4.14.1
Jumble	4	3.3.2	4.4.14.2
Forty-foot drop	5	3.3.3	4.4.14.3
Waterproofness	4	3.3.4	4.4.14.4
Transportation-Vibration	5	3.3.5	4.4.14.5

4.3.3.14.1 Acceptance criteria.-Lot acceptance criteria shall be as specified in 4.2.5.1.

4.3.3.15 Ballistics (destructive).-Eighty (80) fuzes from each lot shall be subjected to the following ballistic test phases:

Phases (ambient temperature)	Quantity	Requirement	Test Method
Target Impact Nonfunction Distance	20	3.4.1 & 2	4.4.15.1
Target Impact Function Distance	10	3.4.1 & 3	4.4.15.2

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4.3.3.15 (Cont'd)

Phases (ambient temperature)	Quantity	Requirement	Test Method
Ground or Water Impact Function			
105MM Howitzer Charge 1	5	3.4.1 & 4	4.4.15.3
105MM Howitzer Charge 7	5	3.4.1 & 4	4.4.15.3
Time-to-Burst			
155MM Howitzer Charge 1	20	3.4.1 & 5	4.4.15.4
105MM Howitzer Charge 7	20	3.4.1 & 5	4.4.15.4

4.3.3.15.1 Acceptance criteria.-The occurrence of any of the following shall be cause for rejection of the lot the ballistic sample represents:

a. A premature function in any phase (see 6.3.1).

b. Function of one or more rounds upon target impact in the Target Impact Nonfunction Distance Phase.

c. Failure of more than one round to function in the Target Impact Function Distance Phase.

d. Failure of more than one round to function in the Ground or Water Impact Function Phase.

e. Any of the following in the Ambient Time-To-Burst Phase at either time setting.

(1) More than one time failure (6.3.2)

(2) More than one outlier-off-time (6.3.3)

(3) Failure to meet the limits specified in Table I (see 3.4.5) by an estimated lot percent defective greater than the appropriate value given in Table II (see 4.2.6.1 and 4.4.15.4.1).

4.3.4 Inspection equipment.-The examinations and tests shall be made using equipment listed in IEL 10520638. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Hardness.-Sample shall be tested in an area which is solidly supported in order to obtain valid readings. The median value of three hardness readings taken on each sample unit shall be used as the individual value.

4.4.2 Bend - Firing arm.-The test shall be performed as specified on Drawing D10521116. The firing arms that have been subjected to the bend test shall be scrapped.

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4.4.3 Load - Support cup.-The test shall be performed using the values specified on Drawing C10521142. Support cups which have been subjected to the applied load test shall be scrapped.

4.4.4 Force - Sintered rotor.-The test shall be performed using the value and direction specified on Drawing F11744012. Rotors which have been subjected to the applied force shall be scrapped.

4.4.5 Compression - Firing pin spring.-The springs shall be compressed for the length of time specified on Drawing C10521132. The springs that have been subjected to the compression test shall be scrapped.

4.4.6 Static load and torque.-Escapement lever assemblies shall be tested prior to insertion of the escapement lever spring. The test shall be performed using the values as specified on the applicable drawing. The load applied to pins shall be in the opposite direction of the assembly of the pins. All samples tested shall be scrapped, except mainspring housing and arbor assemblies on which the lug was subjected to a push test.

4.4.7 Load - Hammer spring.-The required load shall be applied at the specified location parallel to the vertical axis of the assembly as specified on Drawing D10520710.

4.4.8 Load - Setback mechanism assembly.-The test shall be performed using the value specified on Drawing D10551357. The load shall be applied gradually.

4.4.9 Movement subassembly.

4.4.9.1 Kinetic energy of firing pin.-The assembly shall be placed in a fixture and tested for kinetic energy by releasing the firing pin from the cocked position.

4.4.9.2 Dynamic regulation.-The following procedure shall be observed: The movement subassemblies shall be regulated, using the operation procedures furnished with the equipment. The movement subassemblies shall be rotated in a clockwise direction when viewed from the bottom at a constant speed of 6,000 plus or minus 15 rpm. Timekeeping and performance shall be recorded over a period of not less than 100 seconds. No adjustment shall be made on the escapement during the 100 second check run.

4.4.9.3 Spin function.-The wound assembly shall be spun in a clockwise direction when viewed from number one plate end at a constant rpm as specified on Drawing F10520716. The duration of spin shall be for a period of not less than ten seconds. The movement subassembly shall begin functioning immediately at the specified rpm and the lever assembly shall be released during the spin. At the conclusion of the test, the subassembly shall be rewound. The function test shall be performed prior to the nonfunction test.

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4.4.9.4 Spin nonfunction.-The completely wound subassembly (see 6.7) shall be spun in a clockwise direction when viewed from number one plate at a constant rpm as specified on Drawing F10520716. The duration of spin shall be for a period of not less than three (3) seconds. Nonfunction shall be determined by use of a strobe light. This test shall be the last spin test to which the movement is subjected.

4.4.10 Movement assembly.

4.4.10.1 Timing disc torque.-The torque test shall be performed after assembly of the timing disc assembly to the movement and assembly of the lock nut on the arbor. The highest and lowest readings will be taken in one uninterrupted complete revolution of the timing disc for determination of compliance wi the requirements of Drawing F10520715.

4.4.11 Delay arming mechanism assembly.

4.4.11.1 Spin arming less setback assembly.-The unarmed D.A.M. assembly shall be spun in a counterclockwise direction when viewed from movement plate end at a constant rpm as specified on Drawing F10520702. The duration of spin shall be three (3) seconds maximum. At the conclusion of the test, the rotor shall be returned to the unarmed position (see 6.2).

4.4.11.2 Spin, 3000 + 50 r.p.m. with setback assembly.-The unarmed D.A.M. assembly (see 6.2) shall be spun in a counterclockwise direction when viewed from movement plate end at a constant rpm, as specified. The duration of spin shall be three (3) seconds minimum. At the conclusion of the test, the rotor and blocks shall be returned to the unarmed position.

4.4.11.3 Spin nonarming with setback assembly.-The unarmed D.A.M. assembly (see 6.2) shall be spun in a counterclockwise direction when viewed from movement plate end at a constant rpm as specified on Drawing F10520702. The duratio of spin shall be three seconds minimum. Nonarming shall be determined by use of a strobe light. The nonarming test shall be the last spin test to which the D.A.M. assembly is subjected prior to assembly into the fuze.

4.4.12 Detent safety mechanism.

4.4.12.1 Arming and nonarming.-The assembly shall be spun in the counterclockwise direction when viewed from the cover plate at the rpm specified on Drawing F10520250. The arming test shall be performed prior to the nonarming test. Arming and nonarming shall be determined by use of a strobe light.

4.4.13 Fuze assembly.

4.4.13.1 Setting torque.-The test shall be performed using the values as specified on the applicable drawing. The highest and lowest readings will be taken in one uninterrupted complete revolution of the cap for determination of compliance with the requirement of F10520720.

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4.4.14 Environmental test (fuze assemblies less closing cup assemblies).-The procedures used shall be as set forth in the following subparagraphs, as applicable. Where spin testing is required, the nonarming or nonfunction test shall be performed prior to the arming or function test. The components of the disassembled fuzes shall be scrapped. All adverse conditions observed shall be reported to the responsible engineering element.

4.4.14.1 Jolt.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 101.1, except that two arms loaded with fuzes and dummies may be used for the test with the fuze orientations in the following sequency:

a. Nose down
b. Horizontal
c. Nose up

4.4.14.1.1 The fuze shall be disassembled and inspected as follows:

a. Prior to disassembly, check position of "S" line relative to zero line.
b. Remove expansion plugs and check engagement of setting pin with timing disc.
c. Remove head and check detent assembly, primer, and firing pin.
d. Remove D.A.M. assembly and check position of blocks. Manually release blocks and apply push load against rotor through kidney hole (where rotor is found in the unarmed position.)
e. Remove reinforcing ring and closing plug, check relay.
f. Remove movement from body. Check position of safetydisc lines relative to firing arm. Check timing disc torque, position of lever lock slides and for separation of movement plates.
g. Spin detent safety mechanism for nonarming and arming.
h. Disassemble major subassemblies to the extent that all critical parts can be examined, such as gear and pinion assemblies, etc., for security of assembly, broken or bent pivots and teeth, fractured or bent firing arm or shaft or other part which might allow accidental release of firing pin. Check condition of all explosive elements.

4.4.14.2 Jumble.-The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 102.1. The fuze shall be disassembled and inspected as outlined in 4.4.14.1.1.

4.4.14.3 Forty-foot drop.-The test shall be performed in accordance with the procedures specified in MIL-STD-331, Test No. 103. The fuze shall be disassembled and inspected as follows:

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4.4.14.3 (Cont'd)

a. Remove D.A.M. and check position of blocks. Manually release blocks and apply push load against rotor through kidney hole (where rotor is found in the unarmed position).

b. Remove D.A.M. cover plate and check condition of parts (where rotor is found in a partially armed or fully armed position).

4.4.14.4 Waterproofness.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 108 except as provided in 3.3.4. The D.A.M. setback housing and flash hole shall be adequately sealed for the test. The fuze shall be disassembled and inspected as follows:

Remove D.A.M., head, cap and movement assembly. Check all internal surfac for evidence of water entry.

4.4.14.5 Transportation-vibration.- The test shall be performed in accordance with the procedure specified in MIL-STD-331, Test No. 104 with the fuze orientatio in the following sequence:

(a) Parallel to the longitudinal axis, nose up.
(b) First transverse axis to be parallel to a line running through the 100 second marking.
(c) Second transverse axis to be parallel to a line running through the 70 second marking.

4.4.14.5.1 The fuze shall be disassembled and inspected as follows:

(a) Prior to disassembly, check position of "S" line relative to zero line. Check position of rotor through the flash hole.
(b) Remove expansion plugs and check engagement of setting pin with timing disc.
(c) Remove head and check detent assembly, primer, and firing pin. Spin detent assembly for nonarming and arming.
(d) Remove D.A.M. assembly; check position of blocks. Remove setback mechanism assembly and test pin for deflection requirements of Drawing D10551357. Spin D.A.M. assembly for nonarming and arming.
(e) Remove movement from body. Check position of safety disc lines relative to firing arm. Check timing disc torque and position of lever lock slides. Spin movement for nonfunction and function.

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4.4.14.5.1 (Cont'd)

(f) Disassemble major subassemblies to the extent that all critical parts can be examined, such as gear and pinion assemblies, etc., for broken or bent pivots and teeth, fractured or bent firing arm or shaft or other part which might allow accidental release of firing pin. Check condition of all explosive elements.

4.4.15 Ballistics.-The fuze assemblies shall be assembled to suitable projectiles and firing as follows:

4.4.15.1 Target impact nonfunction distance phase.-The sample fuzes shall be fired as specified in 3.4.2 against a suitably supported one inch plywood screen of adequate size under the conditions delineated in 4.2.6 or 4.3.3.15 as appropriate. Observation of impact element function is required.

4.4.15.2 Target impact function distance phase.-The sample fuzes shall be fired as specified in 3.4.3 against a suitably supported one inch plywood screen of adequate size under the conditions delineated in 4.2.6 or 4.3.3.15 as appropriate.

4.4.15.3 Ground or water impact function phase.-The sample fuzes shall be fired as specified in 3.4.4 under the conditions delineated in 4.2.6 or 4.3.3.15 as appropriate.

4.4.15.4 Time-to-burst phase.-The sample fuzes shall be fired as specified in 3.4.5 under the conditions delineated in 4.2.6 or 4.3.3.15 as appropriate. Time-to-burst shall be measured to the nearest millisecond by using two independent fuze chronograph systems. Time-to-burst shall also be measured by using two electric clocks or stop watches. All measurements shall be recorded (see 6.3.3.2).

4.4.15.4.1 Lot percent defective.-The time values obtained by the chronograph systems shall be used to calculate the estimate of lot percent defective. Where both chronograph systems provide a time value, the average shall be taken as the individual fuze value. If one chronograph system fails to provide a time value, the one time value obtained from the functioning chronograph system shall be used as the individual fuze value. Individual fuze values shall be rounded off to the second decimal place using the method prescribed by the American Society for Testing Materials. The estimate of lot percent defective shall be calculated using the method prescribed in MIL-STD-414, Example B3. For sample sizes 16 through 19 the estimated lot percent defective value shall be determined by interpolation between the values given in Table B5, MIL-STD-414 for sample sizes 15 and 20. All appropriate time values obtained for each time-to-burst phase shall be used in the initial calculation of the sample average ($\bar{x}$) and standard deviation (s). The standard deviation value obtained shall then be used to determine statistical outliers (non-normal values) by using the criteria $\bar{x}$ plus or minus 2.88 times the standard deviation. A value judged to be an outlier shall be excluded and the remaining values used to recalculate $\bar{x}$ and s. A minimum of 15 statistically normal values are required for each time-to-burst phase (see 6.3.3.1).

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4.4.15.5 Excess pressure phase.-The sample fuzes shall be fired as specified in 3.4.6 under the conditions delineated in 4.2.6. Time-to-burst shall be measured to the nearest millisecond, using two independent timing devices. The data shall be recorded for informational purposes.

4.4.15.6 Extreme temperature phase.-The sample fuzes shall be fired as specified in 3.4.7 and 3.4.8 under the conditions delineated in 4.2.6 and 3.4.9. Time-to-burst shall be measured to the nearest millisecond, using two independent timing devices. The data shall be recorded for informational purposes.

4.4.15.6.1 Temperature conditioning (-65°F and +145°F).-The test chamber shall be ambient temperature at the start of the test and the chamber maintained at a temperature of minus 65°F or plus 145°F as applicable, until stabilization of the fuze temperature is reached (see 6.4). After stabilization, the fuze shall be saturated at minus 65°F or plus 145°F as applicable, for one hour, minimu

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and marking.-Preservation, packaging, packing and marking shall be as specified in the contract and in accordance with MIL-P-60412 with the following exception:

5.1.1 Level"B".-Packaging shall be as required for Level "A" except pressure test is not required.

5.2 The ICC marking shall be "Percussion Fuzes - Handle Carefully".

6. NOTES

6.1 Dimensions.

6.1.1 Compliance with tests (proving ground, ballistic performance, and shop) required by the specification and drawings is mandatory. The contractor will not assume, nor does the Government quarantee, that all possible combinations permitte by the tolerance limits of the specifications and drawings will consistently satisfy the test requirements. Therefore, the manufacturer is obligated to choose those combinations of tolerances and fits within the limits of the specifications and drawings that best suit his process needs and still satisfy the testing requirements.

6.1.2 Dimensions which are specifically referred to in the classification of defects, as listed in Section 4, are mandatory. Pertaining to other dimensions, i is recognized that current mass production methods may occasionally produce parts which are not in full compliance with all dimensional requirements, but which may provide adequate fuze performance. Therefore, the contractor,at his own risk, may utilize these parts provided there is no adverse effect on fuze functionability. The relaxation of requirements applies to "after the fact" deviations only, to dimensions only, and does not constitute authority to establish new dimensions or to relax efforts to achieve the required sizes within the specified limits. The

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6.1.2 (Cont'd)

procedure for accomplishing dimensional changes is available under the established engineering change system. Nothing above is to be construed as relieving the contractor of any responsibility in connection with performance of the fuze.

6.1.3 The contractor shall maintain records showing deviations from dimensional requirements, the extent of such deviations, the amount of material involved, and the date the material is introduced into the assembly line. Copies of these records shall be forwarded to the agency assigned custody of the technical data.

6.2 Armed/unarmed delay arming mechanism.

6.2.1 Armed. Rotor is locked by the spring loaded detent in the position in which the detonator is fully visible through the flash hole.

6.2.2 Unarmed. Rotor fully engaged by both spring loaded blocks and the setback pin preventing rotor movement.

6.3 Ballistic parameters.

6.3.1 Premature function. A premature function shall be defined as the occurrence of any of the following:

a. The round detonates in the bore of the gun.
b. The round detonates before 45 seconds when set at 60 seconds or before 1.9 seconds when set at 3 seconds.
c. The round detonates before impact when set on safe "S" in the Target Impact Nonfunction Distance Phase, Target Impact Function Distance Phase, or the Ground or Water Impact Function Phase.

6.3.2 Time Failure. A time failure is considered to have occurred if the round fails to detonate during flight.

6.3.3 Outlier-offtime. An outlier-offtime is a statistical outlier (non-normal value) which falls beyond the appropriate time limits specified in Table I.

6.3.3.1 Where the number of statistically normal time values obtained in either time-to-burst phase is less than 15 and the lot is not rejected because of other reasons, the disposition of the lot shall be referred to the responsible procuring activity. The Government reserves the right to require the contractor to submit additional test samples. The number of additional test samples shall be sufficient to obtain a combined total of 20 values in order that the occurrence of outliers will not reduce the number of normal values to less than the minimum required. All normal values obtained in the combined tests shall be used to determine lot acceptance or rejection. Attribute characteristics failures, e.g., duds, occurring in the additional samples shall not be used to reject the lot.

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6.3.3.2 Electric clock and stop watch time values. Where only electric clock or stop watch time measurements are obtained and these values indicate a premature function or outlier-offtime which affects lot disposition, the lot shall be referred to the responsible procuring activity.

6.3.3.3 Defective fuzes. Where sample fuzes are found to be defective as received at the ballistic test facility, all tests of the lot sample shall be held in abeyance pending investigation by the responsible procuring activity.

6.3.4 Muzzle velocities mentioned in this specification are nominal only and are given for the guidance of the proving ground in selecting appropriate weapons and rounds.

6.4 Temperature stabilization of fuzes may be checked by a thermal measuring device in good thermal contact with the largest centrally located internal mass. A simulated fuze may be used for this purpose (see 3.4.9).

6.5 Data cards. Data card information shall be as specified in MIL-STD-1167.

6.5. Adequate seating. The gear or pallet shall be seated on at least one portion of the pinion or shaft shoulder and be within the drawing perpendicularity requirements.

6.7 Completely wound movement. A completely wound movement is one which has been wound to the stop and had gear train backlash removed in accordance with the drawing.

6.8 Statement of Safety Requirements.

6.8.1 Safety Limitations and Precautions Relating to Production Equipment and Facilities:

Assembly of explosive elements into fuzes shall be performed under protective hoods with static electricity controls. Protective barriers should be used when performing the MIL-Standard tests.

6.8.2 Hazards Presented by Materials:

Possible injury (burns or dismemberment) to limbs and body from flash or detonation of explosive elements.

6.8.3 Known hazards in Hazardous Areas:

All areas in which explosive elements are used in assembly operations, or stored.

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6.8.4 Specific Instructions Relative to Manufacture, Handling, Storage and Shipment of Material:

The number of persons exposed to explosive operations and the quantities of explosive materials should at all times be the minimum consistent with safety and the efficiency of operation. Signs setting forth personnel and explosive limits also should be posted. Prior to starting operations including any of the hazardous materials or items described herein, a standard operating procedure should be developed and approved by the contractor outlining the specific requirements to safely perform the particular operation. All personnel concerned should be cognizant of SOP contents. Applicable portions of the approved procedure should be conspicuously posted convenient to all stations involved in the operation for guidance of all personnel.

6.8.5 Explosive Hazard Classification of Material:

Assembled fuzes - Class 1, Detonators in Storage - Class 7.

6.9 Ordering data. Procurement documents should specify the following:

Title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1390-A202

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-F-60612C(MU) Fuze, MTSQ, M548 Assembly

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRAC

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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